

Nkosinathi Khumalo
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Campus Marketplace - Uni Square
Software Design Project

Sprint 1 Individual retrospective:

Contribution to the sprint:

During this sprint, my primary focus was on the backend of the project, with some contribution to the front-end design.

Back-end:

- I was responsible for setting up the Express server structure, configuring the Supabase client in `server/src/config/supabaseClient.js` which included writing the `validateUniversityEmail` function, which enforces that only Wits university email domains are accepted during sign-in, and the `getRoleRedirect` function, which redirects users to the correct dashboard based on their role after authentication. And then looked on the listings and how users can query listing
- I also implemented the listings feature for the marketplace. This involved creating `server/src/services/listingService.js` with the `getActiveListings` and `createListing` functions, `server/src/services/listingValidator.js` to validate listing inputs before insertion into the database, and `server/src/routes/listings.js` to expose the `GET /api/listings` and `POST /api/listings` endpoints.
- I also implemented the unit tests for the above functions using Jest for Test Driven Development.

Front End:

- On the frontend side, I contributed to the design and implementation of the Student Dashboard page, building it as a component-based React page using Tailwind CSS and semantic HTML.
- I also contributed to the UI of the landing page by designing it on Figma along with my front end members.

Challenges i faced and what didn't go well:

The most challenging aspect of this sprint was getting comfortable with Jest for unit testing. Writing tests before the implementation, following test-driven development principles was a new workflow for me and took longer than expected to get right. Setting up the React project with Vite and ensuring Tailwind was correctly configured also consumed more time than I had anticipated, which affected how much I could deliver in the early days of the sprint. I wanted to make the backend more responsive for this sprint but unfortunately I couldn't find the time.

What i have learnt:

I gained a better understanding of the need for teamwork, collaboration and synchronization. I learnt how the Git workflow works, and the need to have branches and merging branches.

What i want to change for the next sprint:

If I were to approach this sprint again, I would start my assigned tasks much earlier. I underestimated how long the setup phases , particularly configuring the testing environment and the CI pipeline would take. Starting earlier would have given me more time to refine my implementations and write more comprehensive tests rather than rushing toward the deadline.

Commit History:

```
Nkosinathi@Nathi-s-Laptop MINGW64 ~/Desktop/campus-marketplace (main)
$ git log --author="Nathi006" --all
commit 0a641fcac8061b3770f06aa5e685c9706ef4492f (origin/feature/testing, feature/testing)
Author: Nathi006 <khumalogiven08@gmail.com>
Date: Mon Apr 13 14:53:04 2026 +0200

    update: added test coverage, student dashboard

commit 9f73f545c776adb88da5c0e0143caa88d97e21a7 (origin/feature/Student-Dashboard, feature/Student-Dashboard)
Author: Nathi006 <khumalogiven08@gmail.com>
Date: Sun Apr 12 05:24:25 2026 +0200

    Added mock Student Dashboard

commit b3fa75bae5bac147bd4097135cc6b61b44191a2d
Merge: 31d372e 5e34c3c
Author: Nathi006 <khumalogiven08@gmail.com>
Date: Sun Apr 12 00:04:22 2026 +0200

    fix: added unit testing for user stories 2&3

commit 31d372e02872fc9a413545955ce84c3a50d58dc2
Author: Nathi006 <khumalogiven08@gmail.com>
Date: Sat Apr 11 23:53:02 2026 +0200

    fix: Added unit testing for user stories 2&3

commit 0e56a7b26ff1868d68ea32ac06e3bb9969123f97
Author: Nathi006 <khumalogiven08@gmail.com>
Date: Sat Apr 11 21:00:54 2026 +0200

    ci: add Jest coverage reporting to CI pipeline
```

```
commit 4dc85c7bf49ec6915a8550e2c7c3eb8624ccf076
Author: Nathi006 <khumalogiven08@gmail.com>
Date: Mon Apr 6 12:24:16 2026 +0200

    fix: Added jest dependency on the server side for unit testing

commit 62a8c32a52b44cbc8f42b80aac7d7c637741799e (origin/feature/user-authorization, feature/user-authorization)
Author: Nathi006 <khumalogiven08@gmail.com>
Date: Mon Apr 6 04:34:35 2026 +0200

    feat: implemented supabaseclient and unit testing for user-story 1 - signing in with student email
(END)
```